

VPC TASK - 2

1. Create one VPC, with 1 public subnet and 1 private subnet.

Go to:

VPC → Your VPCs → Create VPC

VPC → Subnets → Create subnet → Pvt-Sub & Pub-Sub

VPC → Internet Gateways → Create internet gateway

Actions → Attach to VPC

VPC → Route Tables → Create route table

Routes → Edit routes → Add route

Public-RT → Subnet associations → Edit subnet associations.--> Add Pub-Sub

VPC → Subnets → Public-Subnet → Actions → Edit subnet settings

Enable VPC peering for cross-region → Enable auto-assign public IPv4 address.

VPC → Route Tables → Create route table

Associate → Private-Subnet → with → Private-RT

Your VPCs

VPCs

VPC encryption controls

Your VPCs (1/2) [Info](#)

Last updated 5 minutes ago

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR
☐	-	vpc-0fff3eb0730a112f0	Available	-	-	Off	172.31.16.0/24
☒	Devops-VPC	vpc-0e4de86deb65c07af	Available	-	-	Off	10.0.2.0/24

Subnets (8) [Info](#)

Last updated less than a minute ago

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	-	subnet-09c2ed21e707a07c9	Available	vpc-0fff3eb0730a112f0	Off	172.31.16.0/24
☐	-	subnet-0a7a2d2ffa74a38af	Available	vpc-0fff3eb0730a112f0	Off	172.31.16.0/24
☐	-	subnet-09e9516764dcfc6b3	Available	vpc-0fff3eb0730a112f0	Off	172.31.16.0/24
☐	-	subnet-0f5e6991df2c26cf9	Available	vpc-0fff3eb0730a112f0	Off	172.31.16.0/24
☐	Pub-SUB	subnet-025628b06156a6291	Available	vpc-0e4de86deb65c07af Devops-VPC	Off	10.0.1.0/24
☐	PVT-SUB	subnet-0b36713036cb0778a	Available	vpc-0e4de86deb65c07af Devops-VPC	Off	10.0.2.0/24

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Updated routes for rtb-00a055aa173d046c2 / PUB-RT successfully
Details

rtb-00a055aa173d046c2 / PUB-RTAct

DetailsInfo

Route table ID
rtb-00a055aa173d046c2

VPC
vpc-0e4de86deb65c07af | Devops-VPC

Main
No

Owner ID
567487919764

Explicit subnet associations
-

Edge associations
-

RoutesSubnet associationsEdge associationsRoute propagationTagsRelated resources

Routes (2)BothEdit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-02b089d01ecc85ed6	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

VPCRoute tablesrtb-00a055aa173d046c2Edit routes

Edit routes: rtb-00a055aa173d046c2 / PUB-RT

Destination
10.0.0.0/16

Target
local

Status
Active

Propagated
No

Route Origin
CreateRouteTable

0.0.0.0/0

local

-

No

CreateRoute

0.0.0.0/0

Internet Gateway

-

No

CreateRoute

igw-02b089d01ecc85ed6

Use: "igw-02b089d01ecc85ed6"

igw-02b089d01ecc85ed6 (DevOps-IGW)

Add routeCancel

VPCSubnetssubnet-025628b06156a6291Edit subnet settings

Edit subnet settingsInfo

Subnet

Subnet ID
subnet-025628b06156a6291

Name
Pub-SUB

Auto-assign IP settingsInfo

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

Enable auto-assign public IPv4 addressInfo

Enable auto-assign customer-owned IPv4 addressInfo
Option disabled because no customer owned pools found.

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✓ You have successfully updated subnet associations for rtb-0d43476a40e5e55e8 / PVT-RT.

Route tables (1/4) [Info](#) Last updated 2 minutes ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	-	rtb-0dfec03c57111fe1a	-	-	Yes	vpc-0e4de86deb65c07af Devo...	567487919...
☒	PVT-RT	rtb-0d43476a40e5e55e8	subnet-0b36713036cb07...	-	No	vpc-0e4de86deb65c07af Devo...	567487919...

rtb-0d43476a40e5e55e8 / PVT-RT

Explicit subnet associations (1) [Edit subnet associations](#)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
PVT-SUB	subnet-0b36713036cb0778a	10.0.2.0/24	-

Route tables (1/4) [Info](#) Last updated 3 minutes ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	-	rtb-0aaac6a29dc4a9c9f	-	-	Yes	vpc-0fff3eb0730a112f0	567487919...
☒	PUB-RT	rtb-00a055aa173d046c2	subnet-025628b06156a6...	-	No	vpc-0e4de86deb65c07af Devo...	567487919...
☐	-	rtb-0dfec03c57111fe1a	-	-	Yes	vpc-0e4de86deb65c07af Devo...	567487919...

rtb-00a055aa173d046c2 / PUB-RT

[Details](#) | [Routes](#) | [Subnet associations](#) | [Edge associations](#) | [Route propagation](#) | [Tags](#) | [Related resources](#)

Explicit subnet associations (1) [Edit subnet associations](#)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Pub-SUB	subnet-025628b06156a6291	10.0.1.0/24	-

2. Enable VPC peering for cross-account (you can collaborate with your friend to do this task).

Create VPC Peering

VPC → Peering connections → Create peering connection

Name: East-to-West-Peering

Requester VPC: DevOps-VPC

Account: My account

Region: Another Region

Region: us-west-2

Acceptor VPC: Default VPC

Accept Peering

Switch to: us-west-2

Go: VPC → Peering connections

You should see: Pending Acceptance

Select it: Actions → Accept request

Add Routes

VPC-1 route table Go to:

us-east-1 → Route Tables → Public-RT

Destination: 172.31.0.0/16

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Target:

→Peering Connection: pcx-xxxxxxx

VPC-2 Route table Default

Destination:

10.0.0.0/16

Target:

→Peering Connection: pcx-xxxxxxx

IN MY case It din/t Worked For me so For this Testing purpose

in Sccurity groups I opened all traffic 0.0.0.0/0 = its connected Succesfully

Security Groups → All Traffic → 0.0.0.0/0

Instead Of Opening all ports I tried this its Worked

Security Groups → All Traffic → I entered N.Virgnaia→cidr 10.0.0.0/16 = it's worked

The screenshot displays the AWS Management Console interface. At the top, the navigation bar shows the user is in the 'United States (Oregon)' region. The main content area is divided into two sections: 'Your VPCs' and 'Subnets (1)'.

In the 'Your VPCs' section, there is a table listing VPCs. The table has columns for Name, VPC ID, State, Encryption controls, Encryption control, Block Public Access, and IPv4 CIDR. Two VPCs are listed: one with ID 'vpc-0cab3cbe54dfac3cc' and another named 'DevOps-West-VPC' with ID 'vpc-07985ad6ed323725a'. The 'DevOps-West-VPC' is selected.

Below the table, the details for 'vpc-07985ad6ed323725a / DevOps-West-VPC' are shown. The 'Details' tab is active, displaying information such as VPC ID, State (Available), Block Public Access (Off), DNS hostnames (Disabled), DNS resolution (Enabled), Tenancy (default), DHCP option set, and Main route table.

A green notification banner at the bottom of the VPC section states: 'You have successfully created 1 subnet: subnet-0a6257eb5b4875f25'.

The 'Subnets (1)' section below shows a table with columns for Name, Subnet ID, State, VPC, Block Public Access, IPv4 CIDR, and IPv6 CIDR. One subnet is listed: 'PUB-WEST-SUB' with ID 'subnet-0a6257eb5b4875f25', which is associated with the 'vpc-07985ad6ed323725a' VPC and has an IPv4 CIDR of '10.1.1.0/24'.

The interface includes search bars, filter buttons, and pagination controls throughout.

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pcx-0225e65f121454e2f

A VPC peering connection **pcx-0225e65f121454e2f** / East-to-West-Peering has been requested.

Remember to change your region to us-west-2 to accept the peering connection.

Actions

pcx-0225e65f121454e2f / East-to-West-Peering

Actions

Details

Requester owner ID

567487919764

Peering connection ID

pcx-0225e65f121454e2f

Status

Initiating Request to 567487919764

Expiration time

Sunday, 6 September 2026 at 16:30:21 GMT+5:30

Accepter owner ID

567487919764

Requester VPC

vpc-0e4de86deb65c07af / Devops-VPC

Requester CIDRs

10.0.0.0/16

Requester Region

N. Virginia (us-east-1)

VPC Peering connection ARN

arn:aws:ec2:us-east-1:567487919764:vpc-peering-connection/pcx-0225e65f121454e2f

Accepter VPC

vpc-07985ad6ed323725a

Accepter CIDRs

-

Accepter Region

Oregon (us-west-2)

DNS

Route tables

Tags

DNS settings

Requester VPC (vpc-0e4de86deb65c07af / Devops-VPC)

Edit DNS settings

[Alt+S] Ask Amazon Q

United States (Oregon)

Sunithmac (567487919764)

Peering connections (1)

Find peering connections by attribute or tag

Actions

Create peering connection

Name	Peering connection ID	Status	Requester VPC	Accepter VPC	Requester CIDRs	Accepter CIDRs
-	pcx-0225e65f121454e2f	Pending acceptance	vpc-0e4de86deb65c07af	vpc-07985ad6ed323725a / Dev...	10.0.0.0/16	-

Peering connections (1/1)

Find peering connections by attribute or tag

Actions

Create peering connection

Name	Peering connection ID	Status	Requester VPC	Accepter VPC	Requester CIDRs	Accepter CIDRs
-	pcx-0225e65f121454e2f	Pending acceptance	vpc-0e4de86deb65c07af	vpc-07985ad6ed323725a / Dev...	10.0.0.0/16	-

pcx-0225e65f121454e2f

Details

DNS

Route tables

Tags

Details

Requester owner ID

567487919764

Peering connection ID

pcx-0225e65f121454e2f

Status

Pending Acceptance by 567487919764

Expiration time

-

Accepter owner ID

567487919764

Requester VPC

vpc-0e4de86deb65c07af

Requester CIDRs

10.0.0.0/16

Requester Region

-

VPC Peering connection ARN

arn:aws:ec2:us-west-2:567487919764:vpc-peering-connection/pcx-0225e65f121454e2f

Accepter VPC

vpc-07985ad6ed323725a / DevOps-West-VPC

Accepter CIDRs

-

Actions

Create peering connection

View details

Accept request

Reject request

Edit DNS settings

Manage tags

Delete peering connection

Peering connection ID	Status	Requester VPC	Accepter VPC	Requester CIDRs	Accepter CIDRs
pcx-0225e65f121454e2f	Pending acceptance	vpc-0e4de86deb65c07af	vpc-07985ad6ed323725a / Dev...	10.0.0.0/16	-

VPC TASK - 2

pcx-0225e65f121454e2f

Pending acceptance

vpc-0e4de86deb65c07af

vpc-07985ad6ed323725a / DevOps-West-VPC

Accept VPC peering connection request

Info

Are you sure you want to accept this VPC peering connection request? (pcx-0225e65f121454e2f)

Requester VPC

vpc-0e4de86deb65c07af

Requester CIDRs

–

Requester owner ID

567487919764(This account)

Accepter VPC

vpc-07985ad6ed323725a / DevOps-West-VPC

Requester Region

N. Virginia (us-east-1)

Accepter owner ID

567487919764(This account)

Requester CIDRs

10.0.0.0/16

Accepter Region

Oregon (us-west-2)

Cancel

Accept request

Your VPC peering connection (pcx-0225e65f121454e2f) has been established.

To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables.

Modify my route tables now

Peering connections (1)

Info

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC	Accepter VPC	Requester CIDRs	Accepter CIDRs
–	pcx-0225e65f121454e2f	Active	vpc-0e4de86deb65c07af	vpc-07985ad6ed323725a / DevOps-West-VPC	10.0.0.0/16	10.0.0.0/16

aws

Search

[Alt+S]

Ask Amazon Q

VPC

Route tables

rtb-00a055aa173d046c2

Edit routes

Edit routes: rtb-00a055aa173d046c2 / PUB-RT

Destination

10.0.0.0/16

0.0.0.0/0

10.1.0.0/16

Add route

Target

local

local

Internet Gateway

igw-02b089d01ecc85ed6

Peering Connection

pcx-

Use: "pcx-"

pcx-0225e65f121454e2f (East-to-West-Peering)

Status

Active

Active

–

pcx-0225e65f121454e2f (East-to-West-Peering)

VPC TASK - 2

aws [Alt+S] Ask Amazon Q

VPC > Route tables > rtb-0ac5190041f93e4dd > Edit routes

Edit routes: rtb-0ac5190041f93e4dd

Destination	Target	Status	Propagated	Route Origin
10.1.0.0/16	local	Active	No	CreateRouteTable
10.0.0.0/16	Peering Connection	-	No	CreateRoute

pcx-0225e65f121454e2f

Use: "pcx-0225e65f121454e2f"

pcx-0225e65f121454e2f

Add route

```

Last login: Sun Aug 30 11:57:39 2026 from 103.160.27.55
[ec2-user@ip-10-0-1-115 ~]$ ping 16.146.92.99
PING 16.146.92.99 (16.146.92.99) 56(84) bytes of data.
^C
--- 16.146.92.99 ping statistics ---
143 packets transmitted, 0 received, 100% packet loss, time 147715ms

[ec2-user@ip-10-0-1-115 ~]$ ping google.com
PING google.com (192.178.155.113) 56(84) bytes of data.
64 bytes from yuiadrs-in-f113.1e100.net (192.178.155.113): icmp_seq=1 ttl=106 time=2.21 ms
64 bytes from yuiadrs-in-f113.1e100.net (192.178.155.113): icmp_seq=2 ttl=106 time=2.19 ms
64 bytes from yuiadrs-in-f113.1e100.net (192.178.155.113): icmp_seq=3 ttl=106 time=2.20 ms
64 bytes from yuiadrs-in-f113.1e100.net (192.178.155.113): icmp_seq=4 ttl=106 time=2.19 ms
64 bytes from yuiadrs-in-f113.1e100.net (192.178.155.113): icmp_seq=5 ttl=106 time=2.19 ms
64 bytes from yuiadrs-in-f113.1e100.net (192.178.155.113): icmp_seq=6 ttl=106 time=2.19 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 2.185/2.193/2.209/0.008 ms
[ec2-user@ip-10-0-1-115 ~]$ ping 16.146.92.99
PING 16.146.92.99 (16.146.92.99) 56(84) bytes of data.
64 bytes from 16.146.92.99: icmp_seq=47 ttl=117 time=56.7 ms
64 bytes from 16.146.92.99: icmp_seq=48 ttl=117 time=56.7 ms
64 bytes from 16.146.92.99: icmp_seq=49 ttl=117 time=56.7 ms
64 bytes from 16.146.92.99: icmp_seq=50 ttl=117 time=56.7 ms
64 bytes from 16.146.92.99: icmp_seq=51 ttl=117 time=56.8 ms
64 bytes from 16.146.92.99: icmp_seq=52 ttl=117 time=56.7 ms
64 bytes from 16.146.92.99: icmp_seq=53 ttl=117 time=56.7 ms
64 bytes from 16.146.92.99: icmp_seq=54 ttl=117 time=56.8 ms
64 bytes from 16.146.92.99: icmp_seq=55 ttl=117 time=56.7 ms
64 bytes from 16.146.92.99: icmp_seq=56 ttl=117 time=56.7 ms
64 bytes from 16.146.92.99: icmp_seq=57 ttl=117 time=56.7 ms
^C
--- 16.146.92.99 ping statistics ---
57 packets transmitted, 11 received, 80.7018% packet loss, time 57839ms
rtt min/avg/max/mdev = 56.673/56.710/56.802/0.042 ms
[ec2-user@ip-10-0-1-115 ~]$ ping 172.31.43.159
PING 172.31.43.159 (172.31.43.159) 56(84) bytes of data.
64 bytes from 172.31.43.159: icmp_seq=1 ttl=127 time=59.3 ms
64 bytes from 172.31.43.159: icmp_seq=2 ttl=127 time=59.4 ms
64 bytes from 172.31.43.159: icmp_seq=3 ttl=127 time=59.3 ms
64 bytes from 172.31.43.159: icmp_seq=4 ttl=127 time=59.3 ms
64 bytes from 172.31.43.159: icmp_seq=5 ttl=127 time=59.4 ms
64 bytes from 172.31.43.159: icmp_seq=6 ttl=127 time=59.3 ms
^C
--- 172.31.43.159 ping statistics ---
```

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3. Set up a VPC Transit Gateway.

Go to:

VPC → Transit Gateways

Configure:

Name: DevOps-TGW

Description:

DevOps Lab Transit Gateway

Amazon side ASN: 64512

Attach VPC-1

VPC → Transit Gateway Attachments → Create transit gateway attachment

Attachment type: VPC

Transit Gateway: DevOps-TGW

VPC: DevOps-VPC

Select the subnet: Public-Subnet

Attach VPC-2 : Do the same for: in uswest – 2a/Another Region vpc

VPC → Transit Gateway Route Tables → Select your TGW route table.

Added new → transit gateway → wait for until it's available → gateway attachments-

The screenshot displays the AWS Management Console interface. At the top, a green notification bar states: "You successfully created tgw-0b83d20e31d8f50e3 / DevOps-TGW." Below this, a blue informational bar suggests visualizing and monitoring Transit Gateways via the AWS Network Manager. The main content area shows the "Transit gateways (1)" list with a table containing one entry: "DevOps-TGW" with ID "tgw-0b83d20e31d8f50e3" and state "Pending". Below the list, the "Details" page for the attachment "tgw-attach-0386274403d9e801f / oregon west" is shown. The details are organized into a grid:

Details			
Transit gateway attachment ID tgw-attach-0386274403d9e801f	State Pending	Resource type VPC	Association state -
Transit gateway ID tgw-0af14342c94888c8b	Resource owner ID 567487919764	Resource ID vpc-0cab3cbe54dfac3cc	Association route table ID -
Association policy table ID -	Transit gateway owner ID 567487919764	DNS support Enable	Security Group Referencing support Enable
IPv6 support Disable	Appliance Mode support Disable	Subnet IDs 4 Subnets	

At the bottom, there are tabs for "Flow logs" and "Tags".

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Transit gateways (1) Info

Find transit gateway by attribute or tag

☐	Name	Transit gateway ID	Owner ID	State
☐	Oregon gate	tgw-0af14342c94888c8b	567487919764	Available

Transit gateway route tables (1/1) Info

Find transit gateway route table by attribute or tag

☒	Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default propagati...
☒		tgw-rtb-097d275cd3a016353	tgw-0af14342c94888c8b	Available	Yes	Yes

Transit gateway route tables: tgw-rtb-097d275cd3a016353

Details | **Associations** | Propagations | Prefix list references | Routes | Tags

Associations (2) Info

Find association by attribute or tag

☐	Attachment ID	Attachment name	Resource type	Resource ID	State
☐	tgw-attach-00a75d82fdc567772	oregon new	VPC	vpc-03f6652eb6ccd4c51	Associated
☐	tgw-attach-0386274403d9e801f	oregon west	VPC	vpc-0cab3cbe54dfac3cc	Associated

- Set up a VPC Endpoint.
Created a S3 Bucket for Endpoint
VPC → Endpoints
Configure Endpoint
Name: S3-Endpoint
Service category: AWS services
Service:
Find: com.amazonaws.us-east-1.s3
Type: Gateway
Select VPC → DevOps-VPC
Select Route Table → Private-RT
Verify Endpoint → VPC → Endpoints

You should see:

S3-Endpoint

State:

Available

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VPC > Endpoints > Create endpoint

Create endpoint

Info

Create the type of VPC endpoint that supports the service, service network or resource to which you want to connect.

Endpoint settings

Specify a name and select the type of endpoint.

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify. Tags help you find and manage your endpoint.

S3-Endpoint

Type

Info

Select a category

☒ AWS services

Connect to services provided by Amazon with an Interface endpoint, or a Gateway endpoint

☐ PrivateLink Ready partner services

Connect to SaaS services which have AWS Service Ready designation with an Interface endpoint. Uses AWS PrivateLink

☐ AWS Marketplace services

Connect to SaaS services that you have purchased through AWS Ma Interface Endpoint

☐ EC2 Instance Connect Endpoint

An elastic network interface that allows you to connect to resources in a private subnet

☐ Resources

Connect to resources like Amazon Relational Database Services (RDS) with a Resource endpoint. Uses AWS PrivateLink

☐ Service networks

Connect to VPC Lattice service networks with a Service network end PrivateLink

☐ Endpoint services that use NLBs and GWLBs

Find services shared with you by service name. Connect to a Network LoadBalancer (NLB) service with an Interface endpoint or to a Gateway LoadBalancer (GWLB) service with a Gateway Load Balancer endpoint

Showing services available in service region: United States (N. Virginia) (us-east-1)

Services (1/2)

Search

Service Name = com.amazonaws.us-east-1.s3

X

Clear filters

	Service Name	Owner	Type
☒	com.amazonaws.us-east-1.s3	amazon	Gateway
☐	com.amazonaws.us-east-1.s3	amazon	Interface

116b825e72087b4b9

Successfully created VPC endpoint

vpce-016b825e72087b4b9

X

vpce-016b825e72087b4b9 / S3-Endpoint

Actions

Details

Endpoint ID

vpce-016b825e72087b4b9

VPC ID

vpc-0e4de86deb65c07af (Devops-VPC)

DNS record IP type

service-defined

Private DNS specified domains

-

Status

Available

Status message

-

IP address type

ipv4

Creation time

Sunday, 30 August 2026 at 19:30:45 GMT+5:30

Service name

com.amazonaws.us-east-1.s3

Service region

us-east-1

Endpoint type

Gateway

Private DNS names enabled

No

Private DNS preference

-

Route tables

Policy

Tags